The documentation and process conversion measures necessary to comply with this document shall be completed by 26 July 2019.

INCH-POUND

MIL-PRF-19500/276K
27 March 2019
SUPERSEDING
MIL-PRF-19500/276J
28 November 2014

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON,
TYPES 2N2323, 2N2324, 2N2326, 2N2328, AND A VERSIONS,
THROUGH-HOLE AND SURFACE MOUNT PACKAGES,
QUALITY LEVELS JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNPN, silicon, reverse-blocking-triode thyristors. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package outlines are a TO-205AA orTO-205AD (formerly TO-5, TO-39) in accordance with figure 1 and a 3 terminal SMD package (U4) in accordance with figure 2.

1.3 Maximum ratings.

Type	V_{RM}	V_{RM} (non-rep)	V_{FBXM}	I_O (1)	I_{FSM} (2)	V_{GKM}	T_{STG}, T_J	$R_{\theta JA}$ max	$R_{\theta JC}$ max	$R_{\theta JSP}$ max	Barometric pressure
	V (pk)	V (pk)	V (pk)	A dc	A(pk)	V (pk)	°C	°C/W	°C/W	°C/W	mm Hg
2N2323, S, U4	50	75	50 (3)	.22	15	6	-65	175	15	6	
2N2323A, AS, AU4	50	75	50 (4)	.22	15	6	to				
2N2324, S, U4	100	150	100 (3)	.22	15	6	+150				
2N2324A, AS, AU4	100	150	100 (4)	.22	15	6					
2N2326, S, U4	200	300	200 (3)	.22	15	6					
2N2326A, AS, AU4	200	300	200 (4)	.22	15	6					
2N2328, S, U4	300	400	300 (3)	.22	15	6					15
2N2328A, AS, AU4	300	400	300 (4)	.22	15	6					15
2N2329, S, U4	400	500	400 (3)	.22	15	6					15
2N2329A, AS, AU4	400	500	400 (4)	.22	15	6					15

(1) This average forward current is for an ambient temperature of 80°C and 180 electrical degrees of conduction. For other operating conditions see figure 3.
(2) Surge current is non-recurrent. The rate of rise of peak surge current shall not exceed 40 A during the first 5 µs after switching from the "off" (blocking) to the "on" (conducting) state. This is measured from the point where the thyristor voltage has decayed to 90 percent of its initial blocking value.
(3) Gate connected to cathode through 1,000 ohm resistor.
(4) Gate connected to cathode through 2,000 ohm resistor.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics. T_C = +25°C, unless otherwise specified.

Limits	V_{FM} I_{FM} = 4 (pk) t_p = 8.5 ms max	I_{HOX} (1) (2)	V_{GT} (1) (3)	I_{GT} (1) (3)	V_{GT} (2) (4)	I_{GT} (2) (4)
			T_A = -65°C through +125°C		T_A= -65°C through +125°C	
	V	mA dc	V dc	µA dc	V dc	µA dc
Min			0.1		0.1	
Max	2.2	2.0	1.0	350	0.8	75

(1) Gate connected to cathode through 1,000 ohm resistor (for non-"A" suffix device types).
(2) Gate connected to cathode through 2,000 ohm resistor (for "A" suffix device types).
(3) Applicable to all non-"A" suffix device types and packages.
(4) Applicable to all "A" suffix device types and packages.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN" and "JANTX".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "2323", "2324", "2326", "2328" and "2329".

1.5.3 Suffix symbols. The "A" suffix symbol indicates a modified version of the non-suffix device.

	A blank designator identifies that the package is a TO-205AA in accordance with figure 1.
A	The "A" designator indicates a version of the thyristor that have electrical characteristics of the device and are packaged in a TO-205AA in accordance with figure 1.
S	The "S" designator identifies that the package is a TO-205AD in accordance with figure 1.
AS	The "AS" designator indicates a version of the thyristor that have altered electrical characteristics of the device and are packaged in a TO-205AD in accordance with figure 1.
U4	The "U4" designator identifies a 3 terminal bottom pad configuration SMD package (see figure 2).
AU4	The "AU4" designator indicates a version of the thyristor that have altered electrical characteristics of the device and is packaged in a U4 SMD package (see figure 2).

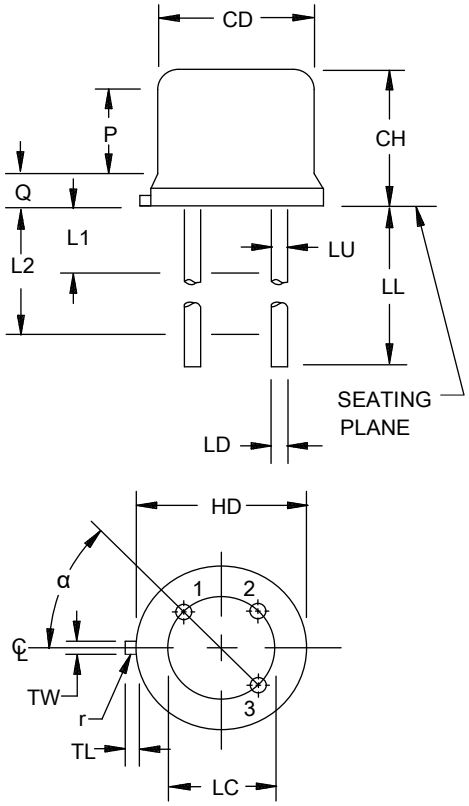
1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	3
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	3
LC	.200 TP		5.08 TP		4
LD	.016	.021	0.41	0.53	5, 6
LL					6, 7
LU	.016	.019	0.41	0.48	5, 6
L1		.050		1.27	5, 6
L2	.250		6.35		5, 6
P	.100		2.54		
Q		.050		1.27	3
TL	.029	.045	0.74	1.14	8
TW	.028	.034	0.71	0.86	9
r		.010		0.25	10
α	45° TP		45° TP		4

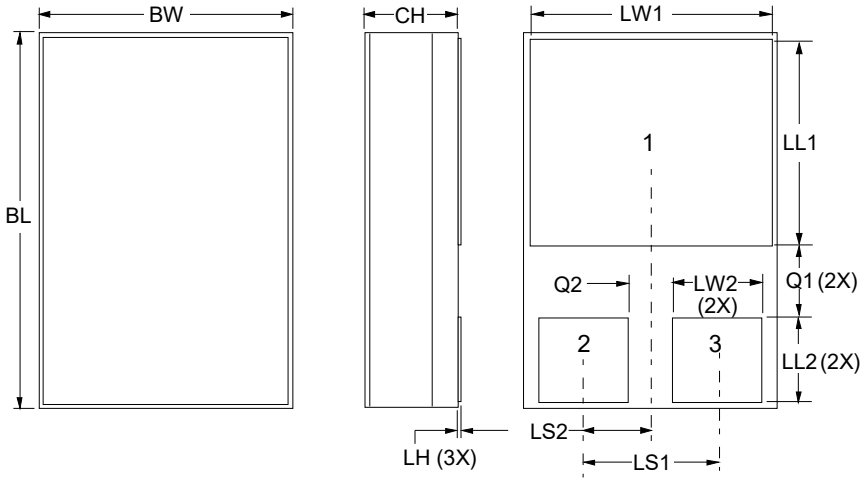


NOTES:

1. Dimension are in inches. Millimeters are given for general information only.
2. Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector. The collector shall be internally connected to the case.
3. Body contour optional within zone defined by dimensions CD, HD, and Q.
4. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
5. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum. Dimension LU applies between dimensions L1 and L2.
6. All three leads.
7. For non-"S" suffix devices (TO-205AA), dimension LL is 1.500 (38.10 mm) minimum, 1.750 (44.45 mm) maximum. For "S" suffix devices (TO-205AD), dimension LL is .500 (12.70 mm) minimum, .750 (19.05 mm) maximum.
8. Dimension TL measured from maximum HD.
9. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-205AA and TO-205AD packages (formerly TO-39, TO-5).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.71
BW	.145	.155	3.68	3.94
CH	.049	.075	1.24	1.90
LH		.020		0.51
LW1	.135	.145	3.43	3.68
LW2	.047	.057	1.19	1.45
LL1	.085	.125	2.16	3.17
LL2	.045	.075	1.14	1.90
LS1	.070	.095	1.78	2.41
LS2	.035	.048	0.89	1.22
Q1	.030	.070	0.76	1.78
Q2	.020	.035	0.51	0.89

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 = anode, terminal 2 =gate, and terminal 3 = cathode.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for 3 terminal SMD package (U4).

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

I_{FBX} DC forward blocking current with specified conditions applied to the gate.
I_{HOX} DC holding current with specified conditions applied to the gate.
I_{RBX} DC reverse blocking current with specified conditions applied to the gate.
R_{GK} External resistance between gate and cathode terminals.
V_{AA} Anode supply voltage (dc).
V_{FBX} Forward blocking voltage, gate connected to the specified reference terminal.
V_{GG} Gate supply voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition specification (see 6.2).

3.4.2 Lead material. Where a choice of lead material is desired, it shall be specified in the acquisition document (see 6.2).

3.4.3 Construction. These devices shall be constructed in a manner and using materials which enable the thyristors to meet the applicable requirements of MIL-PRF-19500 and this specification.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and tables I and II.

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3.7 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANTXV, and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement
	JANTX and JANTXV levels
2	Optional
3a 3b	Required Not applicable
4	Optional
5	Not applicable
7a and 7b	Optional
9	Not applicable
10	Not applicable
11	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}
12	See 4.3.1
(1) 13	Subgroup 2 of table I herein: ΔI_{RBX1} = ±4.0 µA dc; ΔI_{FBX1} = ±4.0 µA dc.
14a and 14b	Required

(1) Devices which exceed the table I limits for this test shall be rejected.

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4.3.1 High temperature blocking. The high temperature blocking test shall be in accordance with figure 4. The test conditions shall be as follows: T_A = 122°C ± 3°C. t = 96 hours minimum. The forward blocking voltage as below:

a. V_{FBX} = 50 V dc for 2N2323, U4, S, A, AU4, AS.

b. V_{FBX} = 100 V dc for 2N2324, U4, S, A, AU4, AS.

c. V_{FBX} = 200 V dc for 2N2326, U4, S, A, AU4, AS.

d. V_{FBX} = 300 V dc for 2N2328, U4, S, A, AU4, AS.

e. V_{FBX} = 400 V dc for 2N2329, U4, S, A, AU4, AS.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E–VIC (small die flow) of MIL-PRF-19500 and herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

Step	Method	Conditions
1	1037	T_A = 75°C ±5°C, I_O = 220 mA dc, see figure 3. No heatsink or forced air on the device shall be permitted. 2,000 cycles, 45 devices, c = 0 R_{GK} = 1 k ohm for 2N2323, S, U4, 2N2324, S, U4, 2N2326, S, U4, 2N2328, S, U4, 2N2329, S, U4. R_{GK} = 2 k ohm for 2N2323A, AS, AU4, 2N2324A, AS, AU4, 2N2326A, AS, AU4, 2N2328A, AS, AU4, 2N2329A, AS, AU4.
2	1037	The intermittent operation life test of step 1 shall be extended to 6,000 cycles. Group B, step 2 shall not be required more than once for any single wafer lot. 45 devices, c = 0.
3	3181	$R_{\theta JA}$ = 175°C C/W maximum (for qualification only). 22 devices, c = 0.

4.4.2.2.1 Sample selection. Samples selected for small die flow group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be selected from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (step 1) may be tested prior to the application of final lead finish.

c. Separate samples may be used for each step.

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4.4.2.2.2 Failures. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during group B small die flow inspection shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Conditions
C2	2036	Condition E. Not applicable for U4 devices.
C5	3181	$R_{\theta JA}$ = 175°C/W; $R_{\theta JC}$ = 15°C/W; $R_{\theta JSP}$ = 6°C/W (for U4 suffix devices only).
C6		Not applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45, c = 0				
Solderability 3/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/	1022	n = 15, c = 0				
Temp cycling 3/	1051	Test condition F, 25 cycles; n = 22, c = 0				
Hermetic seal Fine leak Gross leak	1071	n = 22, c = 0				
Electrical measurements		Table I, subgroup 2				
Bond strength 3/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual	2075	n = 4, c = 0				
Subgroup 2						
Reverse blocking current	4211	DC method, condition B	I_{RBX1}		10	µA dc
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		R_2 = 1 kΩ				
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		R_2 = 2 kΩ				
2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward blocking current	4206	DC method, condition B	I_{FBX1}		10	µA dc
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		R_2 = 1 kΩ				
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		R_2 = 2 kΩ				
2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS		V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc				
Reverse gate current	4219	V_{KG} = 6 V dc	I_{KG}		200	µA dc
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100 Ω				
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		R_e = 1 kΩ	V_{GT1} I_{GT1}	0.35	0.80 200	V dc µA dc
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		R_e = 2 kΩ	V_{GT1} I_{GT1}	0.35	0.60 20	V dc µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +125°C				
Reverse blocking current	4211	DC method, condition B	I_{RBX2}		100	µA dc
		R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4				
		R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4				
2N2323, U4, S, A, AU4, AS		V_R = 50 V dc				
2N2324, U4, S, A, AU4, AS		V_R = 100 V dc				
2N2326, U4, S, A, AU4, AS		V_R = 200 V dc				
2N2328, U4, S, A, AU4, AS		V_R = 300 V dc				
2N2329, U4, S, A, AU4, AS		V_R = 400 V dc				
Forward blocking current	4206	DC method, condition B R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4	I_{FBX2}		100	µA dc
		R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4				
2N2323, U4, S, A, AU4, AS		V_{FBX} = 50 V dc				
2N2324, U4, S, A, AU4, AS		V_{FBX} = 100 V dc				
2N2326, U4, S, A, AU4, AS		V_{FBX} = 200 V dc				
2N2328, U4, S, A, AU4, AS		V_{FBX} = 300 V dc				
2N2329, U4, S, A, AU4, AS		V_{FBX} = 400 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Gate trigger voltage	4221	R_L = 1,000 ohm; R_e = 1 kΩ 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4 R_L = 100 ohm; R_e = 2 kΩ 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4 $V_2 = V_{FBX}$ = 6 V dc 2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS	V_{GT3}	0.1		V dc
Low temperature operation:		T_A = -65°C				
Reverse blocking current	4211	DC method, condition B R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4 R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4	I_{RBX3}		5	µA dc
2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Forward blocking current	4206	DC method, condition B	I_{FBX3}		5	µA dc
		R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4				
		R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4				
2N2323, U4, S, A, AU4, AS		V_{FBX} = 50 V dc				
2N2324, U4, S, A, AU4, AS		V_{FBX} = 100 V dc				
2N2326, U4, S, A, AU4, AS		V_{FBX} = 200 V dc				
2N2328, U4, S, A, AU4, AS		V_{FBX} = 300 V dc				
2N2329, U4, S, A, AU4, AS		V_{FBX} = 400 V dc				
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100 Ω				
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		Re = 1 kΩ	V_{GT2} I_{GT2}		1.0 350	V dc µA dc
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		Re = 2 kΩ	V_{GT2} I_{GT2}		0.8 75	V dc µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Exponential rate of voltage rise	4231	T_A = 125°C, condition B, 50 ohms ≤ R_L ≤ 400 Ω, C = 0.1 to 1.0 µF, repetition rate = 60 pps, test duration = 15 seconds	V_{FBX}			
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		dv/dt = 1.8 V/µs, R_3 = 1 kΩ				
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		dv/dt = 0.7 V/µs, R_3 = 2 kΩ				
2N2323, U4, S, A, AU4, AS		V_{AA} = 50 V dc		47		V dc
2N2324, U4, S, A, AU4, AS		V_{AA} = 100 V dc		95		V dc
2N2326, U4, S, A, AU4, AS		V_{AA} = 200 V dc		190		V dc
2N2328, U4, S, A, AU4, AS		V_{AA} = 300 V dc		285		V dc
2N2329, U4, S, A, AU4, AS		V_{AA} = 400 V dc		380		V dc
Forward on voltage	4226	I_{FM} = 4 A(pk) (pulse), pulse width = 8.5 ms, max; duty cycle = 2 percent max	V_{FM}		2.2	V (pk)
Holding current	4201	Condition B, V_{AA} = 24 V dc max, I_{F1} = 100 mA dc, I_{F2} = 10 mA dc, gate trigger source voltage = 6 V dc, trigger pulse width = 25 µs minimum, R_2 = 330 Ω	I_{HOX}		2.0	mA dc
2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		R_3 = 1 kΩ				
2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		R_3 = 2 kΩ				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, see figure 5; T_A = 122°C, I_{FM} = 15 A, ten surges, 1 per minute, I_O = 0 at rated V_{RWM}, f = 60 Hz				
End-point electrical measurements		See table II, steps 1, 2, 3, and 4				
Subgroup 7						
Non-repetitive peak voltage		See figure 6, ten pulses				
2N2323, U4, S, A, AU4, AS		V_{RM} = 75 V dc				
2N2324, U4, S, A, AU4, AS		V_{RM} = 150 V dc				
2N2326, U4, S, A, AU4, AS		V_{RM} = 300 V dc				
2N2328, U4, S, A, AU4, AS		V_{RM} = 400 V dc				
2N2329, U4, S, A, AU4, AS		V_{RM} = 500 V dc				
End-point electrical measurements		See table II, steps 1, 2, 3, and 4				

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed table I, subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.

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TABLE II. Groups A, B, and C end-point electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse blocking current	4211	DC method, condition B R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4 R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4	I_{RBX1}		10	µA dc
	2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc				
2.	Forward blocking current	4206	DC method, condition B R_2 = 1 kΩ for 2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4 R_2 = 2 kΩ for 2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4	I_{FBX1}		10	µA dc
	2N2323, U4, S, A, AU4, AS 2N2324, U4, S, A, AU4, AS 2N2326, U4, S, A, AU4, AS 2N2328, U4, S, A, AU4, AS 2N2329, U4, S, A, AU4, AS		V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc				
3.	Forward on voltage	4226	I_{FM} = 4A(pk) (pulse), pulse width = 8.5 ms max; duty cycle = 2 percent max	V_{FM}		2.2	V (pk)

See footnotes at end of table.

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TABLE II. Groups A, B, and C end-point electrical measurements - Continued. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
4.	Gate trigger voltage	4221	$V_2 = V_{FBX} = 6$ V dc; $R_L = 100\ \Omega$				
	2N2323, S, U4 2N2324, S, U4 2N2326, S, U4 2N2328, S, U4 2N2329, S, U4		$R_e = 1\ k\Omega$	$V_{GT}1$ I_{GT1}	0.35	0.80 200	V dc µA dc
	2N2323A, AS, AU4 2N2324A, AS, AU4 2N2326A, AS, AU4 2N2328A, AS, AU4 2N2329A, AS, AU4		$R_e = 2\ k\Omega$	V_{GT1} I_{GT1}	0.35	0.60 20	V dc µA dc

1/ The end-point electrical measurements for group A (table E-V of MIL-PRF-19500) are as follows: Subgroups 6 and 7, see table II herein, steps 1, 2, 3, and 4.

2/ The end-point electrical measurements for group B (table E-VIC of MIL-PRF-19500) are as follows:
a. Step 1, see table II herein, steps 1, 2, 3, and 4.
b. Step 2, see table II herein, steps 1, 2, 3, and 4.

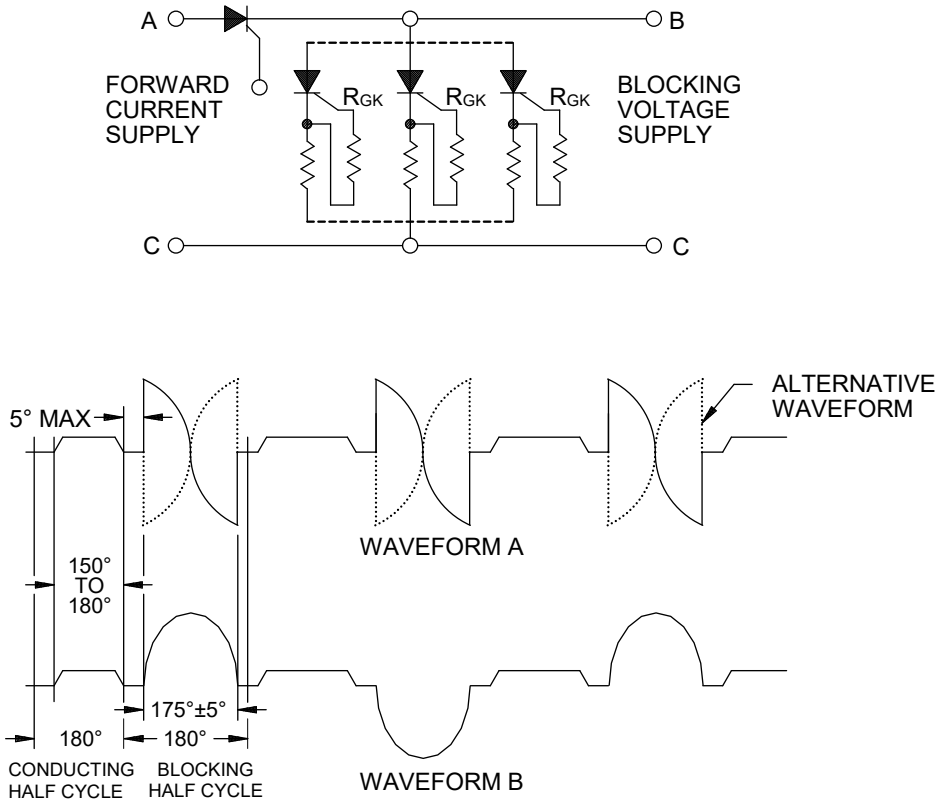
3/ The end-point electrical measurements for group C (table E-VII of MIL-PRF-19500) are as follows:
a. Subgroup 2, see table II herein, steps 1, 2, 3, and 4.
b. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.

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TABLE III. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling or	1051	500 cycles, -65°C to +150°C.	
Thermal shock	1056	100 cycles, test condition B.	
Hermetic seal Fine leak Gross leak	1071		
End-point electrical measurement		See table I, subgroup 2.	
Subgroup 2			
Blocking life	1048	1,000 hours minimum, T_A = 122°C ±3°C, V_{FBX} = 400 V.	
End-point electrical measurements		See table I, subgroup 2.	
Subgroup 4			
Thermal impedance curves			
Subgroup 5			
Barometric pressure (reduced)	1001		
End-point electrical measurements		See table I, subgroup 2.	
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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Test circuit details: The test circuit used must provide one of the waveforms shown. When a large number of devices are to be tested, they may be connected in parallel or series combinations with suitable provisions (resistive, reactive, etc, methods) for proper division of forward current or blocking voltages. Suitable protective elements should be used to isolate defective devices without interrupting the test and to protect the remaining devices on test. The forward current supply shall consist of a low-voltage transformer with a means of adjusting the secondary voltage. It shall supply 22 mA dc average forward current to each device under test. The gate supply shall be in phase with the forward current supply. The average power input shall not be allowed to exceed .01 watt per device.

Operating conditions: The devices shall be subjected to the following operational conditions:

a. Power sources are to be 60 Hz sinusoidal waveform sources.

b. The conduction angle of I_O shall be 150 degrees to 180 degrees.

c. The rated forward or reverse blocking voltage shall be started during the nonconducting half cycle no later than 5 degrees after conduction has ceased. For waveform A, the IRC time constant of the blocking voltage shall be 200 ±100 microseconds.

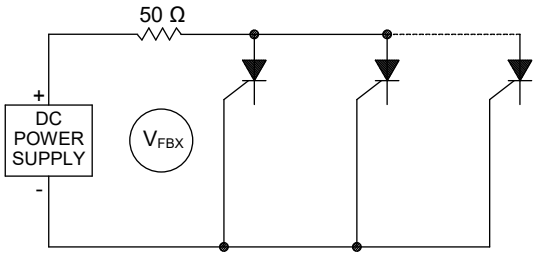
d. For accelerated steady-state operation, blocking voltage shall be equal to or greater than 30 V.

e. The duration of the blocking voltages shall be 175 ±5 degrees.

f. Ambient test temperature shall be 75 ±5°C.

FIGURE 3. Test circuit and waveforms for intermittent operation life test.

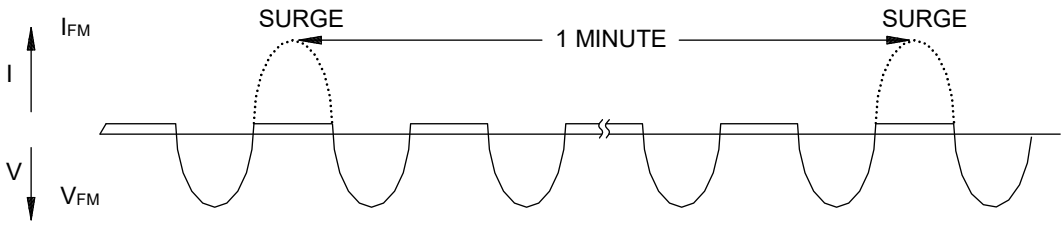
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Procedure: Adjust the power supply to the specified value for the device under test.

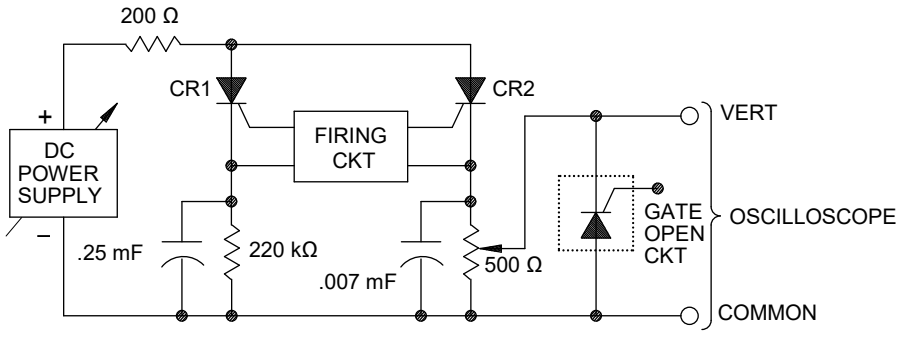
NOTE: Rate of rise of voltage must be limited to values specified in table I.

FIGURE 4. Blocking-voltage test.



Procedure: The above waveform shall be applied to the devices under test. During this test, the gate shall be connected to the anode through a suitable firing resistor and series diode or fired by an equivalent method. The device shall be operated at the specified maximum reverse voltage and specified temperature in a single phase circuit with a 60 Hz supply and a resistive load. The total peak device current during the surge shall be 15 amperes.

FIGURE 5. Surge current waveform.



Procedure: Adjust potentiometer to achieve the specified voltage across the device under test. Timing sequence of firing circuit is: Apply trigger pulse to CR2 once every 15 seconds, apply trigger pulse to CR1, 5.0 ±1.0 milliseconds after firing CR2. The specified number of pulses shall be applied. Alternate circuits shall apply rectangular pulses of 5.0 ±1.0 milliseconds with a maximum rise and fall time of 50 microseconds.

FIGURE 6. Non-repetitive peak reverse voltage test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

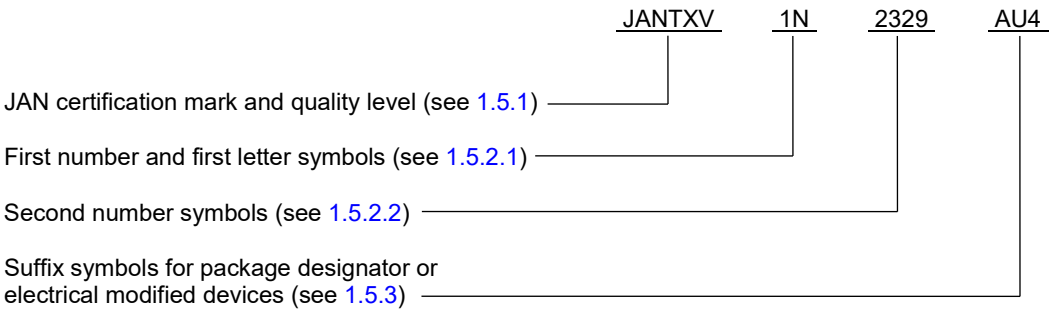
b. Packaging requirements (see 5.1).

c. Lead finish and material (see 3.4.1 and 3.4.2).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs are constructed using the following form.



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6.5 List of PINs.

PINs for devices in a TO-205AA package					
Quality level JAN	Quality level JANTX	Quality level JANTXV	Quality level JAN	Quality level JANTX	Quality level JANTXV
JAN2N2323	JANTX2N2323	JANTXV2N2323	JAN2N2323A	JANTX2N2323A	JANTXV2N2323A
JAN2N2324	JANTX2N2324	JANTXV2N2324	JAN2N2324A	JANTX2N2324A	JANTXV2N2324A
JAN2N2326	JANTX2N2326	JANTXV2N2326	JAN2N2326A	JANTX2N2326A	JANTXV2N2326A
JAN2N2328	JANTX2N2328	JANTXV2N2328	JAN2N2328A	JANTX2N2328A	JANTXV2N2328A
JAN2N2329	JANTX2N2329	JANTXV2N2329	JAN2N2329A	JANTX2N2329A	JANTXV2N2329A
PINs for devices in a TO-205AD package					
JAN2N2323S	JANTX2N2323S	JANTXV2N2323S	JAN2N2323AS	JANTX2N2323AS	JANTXV2N2323AS
JAN2N2324S	JANTX2N2324S	JANTXV2N2324S	JAN2N2324AS	JANTX2N2324AS	JANTXV2N2324AS
JAN2N2326S	JANTX2N2326S	JANTXV2N2326S	JAN2N2326AS	JANTX2N2326AS	JANTXV2N2326AS
JAN2N2328S	JANTX2N2328S	JANTXV2N2328S	JAN2N2328AS	JANTX2N2328AS	JANTXV2N2328AS
JAN2N2329S	JANTX2N2329S	JANTXV2N2329S	JAN2N2329AS	JANTX2N2329AS	JANTXV2N2329AS
PINs for devices in a U4 package					
JAN2N2323U4	JANTX2N2323U4	JANTXV2N2323U4	JAN2N2323AU4	JANTX2N2323AU4	JANTXV2N2323AU4
JAN2N2324U4	JANTX2N2324U4	JANTXV2N2324U4	JAN2N2324AU4	JANTX2N2324AU4	JANTXV2N2324AU4
JAN2N2326U4	JANTX2N2326U4	JANTXV2N2326U4	JAN2N2326AU4	JANTX2N2326AU4	JANTXV2N2326AU4
JAN2N2328U4	JANTX2N2328U4	JANTXV2N2328U4	JAN2N2328AU4	JANTX2N2328AU4	JANTXV2N2328AU4
JAN2N2329U4	JANTX2N2329U4	JANTXV2N2329U4	JAN2N2329AU4	JANTX2N2329AU4	JANTXV2N2329AU4

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693–1642 or DSN 850–6939.

6.7 Changes from previous issue. The margins of this specification are marked with vertical bars to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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